

Emergency Department Patient Flow Analytics Project Plan

Item	Details
Project type	Two-person healthcare analytics project
Team members	Owen and Nimi
Owen focus	Python, exploratory data analysis, machine learning, Streamlit dashboard
Nimi focus	SQL database design, SQL cleaning, KPI queries, reporting views, and healthcare BI questions
Team GitHub link	https://github.com/owendiche-sys/Emergency-Department-Patient-Flow-Analytics-.git
Plan version/date	Version 2.0 - 14 July 2026

Project summary

This project builds an end-to-end Emergency Department (ED) patient flow analytics system using the 2022 NHAMCS Emergency Department Public Use File. The project combines a SQL business intelligence layer, Python analysis, machine learning, an interactive dashboard, and management focused reports. It analyses ED waiting times, arrival patterns, triage acuity, ambulance arrivals, admission outcomes, and patients leaving before care is completed.

The project scope has been narrowed from general hospital patient journey analytics to ED patient flow because the selected dataset contains sampled emergency-department visits and does not include a hospital department field. Therefore, the project will not compare hospital departments or make unsupported department-pressure claims.

Document Control

Version	Date	Author(s)	Changes
1.0	28 June 2026	Owen and Nimi	Initial project plan created.
1.1	28 June 2026	Owen and Nimi	Updated timeline to 01 July - 12 August 2026, moved planning sessions to Sundays, and added explicit business questions.
2.0	14 July 2026	Owen and Nimi	Re-scoped the project to Emergency Department patient flow based on the selected NHAMCS ED dataset. Removed unsupported department and readmission claims, updated KPIs, SQL views, ML features, dashboard pages, and final recommendations.

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1. Project Specifications

1.1 Project Topic

Emergency Department Patient Flow Analytics: SQL KPI Reporting, ML-Based Extended-Wait Prediction and Healthcare Business Storytelling.

The project analyses how patients move through an ED visit episode, from arrival and triage through waiting, care outcomes, admission, observation, discharge, or leaving before care is completed. The final output is an analytics workflow that combines SQL reporting, Python analysis, machine learning, a Streamlit dashboard, and management focused reporting.

1.2 Why Emergency Department?

The focus is the Emergency Department because the selected dataset is the 2022 NHAMCS Emergency Department Public Use File, where each row represents one sampled ED visit. The dataset includes ED relevant variables such as arrival time, waiting time, triage level, ambulance arrival, visit length, admission outcomes, and whether the patient left before being seen or before treatment was complete.

This makes ED patient flow the most valid and evidence-based project focus. Emergency departments are also operationally important because they handle unpredictable arrivals, urgent triage decisions, long-wait risks, ambulance arrivals, hospital admissions, and patients who may leave before care is completed.

1.3 Problem Statement

Emergency departments often experience operational pressure caused by fluctuating arrivals, triage demand, long waiting times, ambulance arrivals, admission bottlenecks, and patients leaving before care is completed. Without a clear analytics system, it can be difficult for hospital managers to identify peak pressure periods, understand which patient groups experience longer waits, monitor key outcomes, and communicate evidence-based recommendations.

This project addresses that issue by developing an ED patient flow analytics system that monitors operational KPIs, answers healthcare business questions, predicts extended-wait risk, and supports practical recommendations for ED management.

1.4 Objectives

- Define healthcare business questions that guide the SQL analysis, dashboard pages, and final recommendations.
- Design a SQL database to store ED visit, timing, triage, outcome, demographic, and operational fields.
- Use SQL to clean the data, create analysis-ready views, and calculate ED operational KPIs.
- Use Python to perform exploratory data analysis and visualise ED patient flow patterns.
- Train and evaluate a machine learning model that predicts whether a patient is likely to experience an extended wait.
- Build an interactive Streamlit dashboard that presents SQL insights, Python visualisations, model results, and recommendations.
- Create executive and business insight reports from a hospital management perspective.
- Document the project clearly so both team members have visible, evidence-based contributions on GitHub.

1.5 Goal / Target

Main goal: To build a complete ED patient flow analytics solution that helps users understand visit patterns, identify waiting-time bottlenecks, monitor ED operational pressure, estimate extended-wait risk, and communicate practical recommendations through a clear SQL-to-Python-to-dashboard pipeline.

1.6 Key Performance Indicators (KPIs)

KPI category	KPI	Definition / success measure
ED flow	Total ED visits	Count of sampled ED visits in the analysis dataset.
ED flow	Visits by arrival period	Visit counts by month, day, and arrival hour.
Wait performance	Average wait time	Mean wait_time_minutes for visits with valid wait times.
Wait performance	Median wait time	Median wait_time_minutes for visits with valid wait times.
Wait performance	2-hour extended-wait rate	Percentage of visits where wait_time_minutes is greater than 120 minutes.
Wait performance	4-hour long-wait rate	Percentage of visits where wait_time_minutes is greater than 240 minutes.

Patient acuity	Wait time by triage level	Average or median wait time grouped by triage urgency.
Arrival mode	Ambulance-arrival rate	Percentage of visits arriving by ambulance.
Outcomes	Admission rate	Percentage of ED visits resulting in admission to hospital.
Outcomes	Left-without-being-seen rate	Percentage of visits where the patient left before being seen.
Outcomes	Left-before-treatment-complete rate	Percentage of visits where the patient left before treatment was complete.
Visit duration	Average and median visit length	Visit length in minutes for visits with valid visit-length values.

1.7 Business Questions

- When are ED visits highest by month, day, and arrival hour?
- What are the average and median ED wait times?
- What percentage of patients wait more than 2 hours or 4 hours?
- Which triage levels and patient groups experience longer waits?
- How do ambulance arrivals differ from non-ambulance arrivals?
- What proportion of ED visits result in admission, discharge, observation, or patients leaving before care is completed?
- Which time periods show the strongest signs of ED operational pressure?
- Can arrival-time information help predict whether a patient is at risk of an extended wait?
- What practical actions could hospital managers take to improve ED flow and reduce long waits?

1.8 Scope Limitations

- The dataset does not include a hospital department field, so the project will not compare departments.
- The dataset does not support readmission analysis, so readmission KPIs and views are excluded.
- The machine learning model is an analytical prototype and not a clinical or operational automation tool.
- Survey weights are available and should be handled carefully when presenting nationally representative estimates.

2. Project Timeline

2.1 Project Tasks, Dates, Deliverables, and Assessments

Task	Owner	Start	End	Deliverable / assessment
Project scope, repo setup, and role confirmation	Owen + Nimi	01 Jul 2026	05 Jul 2026	Project specification, GitHub repository, agreed folder structure.
Dataset selection and ED scope confirmation	Owen + Nimi	01 Jul 2026	05 Jul 2026	NHAMCS ED dataset selected, metadata table, ethics statement, storage plan.
Business questions and KPI mapping	Owen + Nimi	01 Jul 2026	12 Jul 2026	ED business questions mapped to SQL queries, dashboard pages, and reports.
SQL schema design and data loading	Nimi	06 Jul 2026	12 Jul 2026	SQL create tables script, data import script, staging table, final typed ED visits table.
SQL cleaning and KPI queries	Nimi	13 Jul 2026	19 Jul 2026	Cleaning queries, ED KPI queries, and validation checks.
Python EDA and visual analysis	Owen	13 Jul 2026	22 Jul 2026	EDA notebook, summary plots, initial ED flow insights.
SQL views for Python and dashboard	Nimi	20 Jul 2026	26 Jul 2026	ED patient flow, wait KPI, triage, outcomes, and ML feature views.
ML feature engineering and training	Owen	20 Jul 2026	30 Jul 2026	Model notebook/script, model metrics, feature importance.
Dashboard build and integration	Owen + Nimi	20 Jul 2026	03 Aug 2026	Streamlit app with ED KPI pages, SQL insights, ML output, and recommendations.
Executive summary and business insights writing	Owen + Nimi	27 Jul 2026	09 Aug 2026	reports/executive_summary.md and reports/business_insights.md.
Testing and evaluation	Owen + Nimi	03 Aug 2026	11 Aug 2026	Data quality tests, SQL validation, model evaluation, dashboard user checks.
Final review and GitHub cleanup	Owen + Nimi	10 Aug 2026	12 Aug 2026	Final README, reports, dashboard screenshots, completed repository.

2.2 Weekly Milestones

Milestone	Due date	Evidence
Week 1 planning complete	Sunday 05 Jul 2026	Project plan agreed, GitHub repository created, folder structure confirmed, ED scope identified.
Week 2 data and SQL foundation complete	Sunday 12 Jul 2026	NHAMCS ED dataset loaded, SQL schema started/completed, Python and SQL row counts validated.
Week 3 SQL insight layer complete	Sunday 19 Jul 2026	SQL cleaning queries, ED KPI queries, and first SQL views completed.
Week 4 analysis and dashboard structure complete	Sunday 26 Jul 2026	Python EDA visuals completed, dashboard skeleton available, SQL views ready for integration.
Week 5 ML and dashboard draft complete	Sunday 02 Aug 2026	Working ML draft, early model evaluation, dashboard pages populated with SQL and Python outputs.
Week 6 reporting and final dashboard draft complete	Sunday 09 Aug 2026	Executive summary draft, business insights draft, dashboard nearly final, documentation updated.
Final submission ready	Wednesday 12 Aug 2026	Clean GitHub repository, final README, final reports, dashboard screenshots, tested app, and contribution statement.

3. Data Management Plan

3.1 Dataset Overview

Item	Details
Dataset name	2022 NHAMCS Emergency Department Public Use File
Publisher	US Centers for Disease Control and Prevention, National Center for Health Statistics
Source	https://www.cdc.gov/nchs/nhamcs/documentation/index.html
Technical documentation	https://ftp.cdc.gov/pub/Health_Statistics/NCHS/Dataset_Documentation/NHAMCS/doc22-ed-508.pdf
Access date	3 July 2026
Unit of observation	One sampled emergency-department visit
Source data	16,025 visits and 913 variables
Prepared data	16,025 visits and 38 selected variables
Privacy	Public-use microdata with disclosure protections; no direct patient identifiers in selected fields.

3.2 Data Collection and Storage

- The raw CDC/NCHS Stata file is stored locally in data/raw and excluded from Git where appropriate.
- The reproducibly prepared CSV is stored in data/processed for SQL and Python analysis.
- The cleaning scripts record selected fields, missing-value handling, decoded categories, and wait-time flags.
- Dataset source, download date, licence notes, and preprocessing decisions are recorded in README and docs.
- SQL scripts load the processed CSV into a local database, using a staging table before the final typed table.
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3.3 Metadata / Data Dictionary

Field group	Example fields	Use in project
Visit identifiers	visit_id	Primary key and validation checks.
Arrival timing	visit_month, visit_day, arrival_time, arrival_hour	Volume trends, peak arrival analysis, ML features.
Waiting and length	wait_time_minutes, visit_length_minutes	Wait KPIs, long-wait flags, visit duration analysis.
Patient characteristics	age_years, sex, residence_type	Patient-group analysis and candidate ML features.
Acuity and arrival mode	triage_level, arrival_by_ambulance, ambulance_transfer	ED pressure, triage, and arrival-mode analysis.
Clinical context	pain_scale, vital signs, chronic_condition_count, diagnosis_count, procedure_count	EDA and candidate near-arrival ML features where suitable.
Outcomes	admitted_to_hospital, observation outcomes, left_without_being_seen, left_before_treatment_complete	Outcome KPIs and reporting; excluded from arrival-time wait prediction features.
Geography	region, metropolitan_status	Regional and metropolitan-status comparisons.
Targets	extended_wait_2hr_flag, long_wait_4hr_flag	Machine learning target candidates and dashboard wait-rate KPIs.

Document Control Using GitHub

Control item	Plan
Repository	https://github.com/owendiche-sys/Emergency-Department-Patient-Flow-Analytics-.git
Branching system	main for stable work; feature/sql-nimi for SQL work; feature/python-owen for Python, ML, and dashboard work.
Pull requests	Each major change should be reviewed before merging into main.

Commit messages	Use clear messages such as sql: create schema, sql: add ED KPI views, python: add EDA notebook, dashboard: add ML page, reports: add business insights.
File naming	Use numbered files such as 01_create_tables.sql, 03_cleaning_queries.sql, 01_python_eda.ipynb, and 02_ml_long_wait_prediction.ipynb.
Versioning	Use tags or release notes for final versions, for example v1.0-final.

3.4 README File

- Project overview, ED-focused problem statement, and justification for the ED scope.
- Team members and contribution summary.
- Business questions and KPI definitions.
- Dataset source, licence, data dictionary, and limitations.
- How to create the SQL database and run the SQL scripts.
- How to run the Python notebooks and Streamlit dashboard.
- Screenshots of the dashboard.
- Model results, class imbalance notes, and limitations.
- Links to reports/executive_summary.md and reports/business_insights.md.
- Final insights and recommendations.
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3.5 Security and Storage

- No direct patient identifiers such as names, addresses, phone numbers, emails, NHS numbers, or exact dates of birth will be stored in the repository.
- Any local database credentials will be stored in a .env file and excluded from GitHub using .gitignore.
- Raw data will only be committed if the dataset licence allows redistribution.
- The project will be presented as an educational analytics project, not a clinical decision-making tool.

3.6 Ethical Statement

Ethical question	Project response
What data is being used, and is it personal?	The project uses CDC/NCHS public-use ED visit microdata. Direct patient identifiers are not included in the selected fields.
How will privacy and confidentiality be protected?	No names, contact details, exact addresses, or secret credentials will be included. Sensitive files and credentials will be excluded from GitHub.
Could the analysis create bias or unfair outcomes?	The team will avoid using the model as a clinical decision tool. Demographic findings will be interpreted carefully and with limitations.
How will the outputs be used responsibly?	The dashboard and model will be used for learning, portfolio demonstration, and operational insight. Recommendations will be framed as decision-support only.

4. Pipeline Workflow

4.1 Workflow Stages

Stage	Owner	Output
Raw data collection	Owen + Nimi	NHAMCS ED dataset stored or referenced with licence notes.
Business question mapping	Owen + Nimi	ED business questions linked to SQL KPIs, dashboard pages, and final reports.
Database design	Nimi	Schema for ED visits, timing, triage, outcomes, geography, and candidate ML fields.
SQL cleaning	Nimi	Cleaned tables, standardised categories, wait flags, and validation checks.
SQL KPI reporting	Nimi	Views such as vw_ed_patient_flow_summary, vw_ed_wait_kpis, and vw_ml_wait_features.
Python EDA	Owen	Visual analysis of ED waiting times, patient flow, triage, arrivals, and outcomes.
Machine learning	Owen	Extended-wait prediction model and evaluation metrics.
Dashboard integration	Owen + Nimi	Streamlit dashboard using SQL outputs and Python results.
Reporting and storytelling	Owen + Nimi	README, executive summary, business insights, screenshots, and final recommendations.

4.2 Proposed Repository Structure

emergency-department-patient-flow-analytics/

```
|-- data/
| |-- raw/
| |-- processed/
|-- docs/
| |-- Project Plans/
|-- sql/
| |-- 01_create_tables.sql
| |-- 02_load_data.sql
| |-- 03_cleaning_queries.sql
| |-- 04_kpi_queries.sql
| |-- 05_views_for_python.sql
|-- notebooks/
| |-- 01_python_eda.ipynb
| |-- 02_ml_long_wait_prediction.ipynb
|-- src/
|-- dashboard/
| |-- app.py
|-- outputs/
| |-- figures/
| |-- models/
|-- reports/
| |-- executive_summary.md
| |-- business_insights.md
|-- README.md
|-- requirements.txt
```

4.3 SQL Views for Python and Dashboard

SQL view	Purpose
vw_ed_patient_flow_summary	Summarises ED visit volume, wait performance, admission outcomes, and visit-length patterns.
vw_ed_wait_kpis	Provides average wait, median wait, 2-hour extended-wait rate, and 4-hour long-wait rate.
vw_ed_arrival_patterns	Aggregates volume and wait metrics by month, day, and arrival hour.
vw_ed_triage_flow	Summarises visits, waits, ambulance arrivals, and outcomes by triage level.
vw_ed_outcomes	Summarises admission, observation, discharge-related outcomes, left-without-being-seen, and left-before-treatment-complete indicators.
vw_business_question_metrics	Connects management questions to KPI outputs used in the dashboard and business insights report.

vw_ml_wait_features	Provides modelling features for Python, excluding post-arrival outcome leakage.
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4.4 Dashboard Pages

- Overview: headline ED KPIs and filters.
- ED Patient Flow: volume by month, day, and arrival hour.
- Waiting Time Analysis: wait distribution, median wait, 2-hour and 4-hour rates.
- Triage and Acuity: wait patterns by triage level and ambulance arrival.
- Outcomes: admission, observation, left-without-being-seen, and treatment-completion outcomes.
- SQL Insights: selected SQL KPI outputs and query-backed findings.
- ML Prediction: extended-wait model performance, features, and limitations.
- Business Recommendations: management-focused interpretation and next steps.

5. Quality Assurance and Testing Plan

Test area	Checks	Owner
Data quality	Check duplicates, missing values, invalid times, negative waiting times, impossible values, and inconsistent categories.	Nimi + Owen
SQL validation	Compare SQL row counts, wait-time coverage, flag totals, and selected KPIs with Python validation outputs.	Nimi
Business question validation	Check that each business question is answered by at least one SQL query, dashboard chart, or report section.	Owen + Nimi
Python validation	Check dataframe shapes, data types, null values, and consistency between SQL exports and Python inputs.	Owen
ML evaluation	Use train/test split, compare baseline and ML models, and report confusion matrix and classification metrics.	Owen
Leakage review	Confirm post-arrival outcomes such as admission status, visit length, and leaving outcomes are not used for arrival-time wait prediction.	Owen + Nimi
Dashboard testing	Test filters, charts, KPI cards, mobile/desktop layout, and error messages for missing data.	Owen + Nimi
Documentation review	Check README instructions, screenshots, file links, reports, limitations, and contribution summary.	Owen + Nimi

5.1 Model Evaluation Criteria

- The model should be compared against a simple baseline, such as predicting the majority class.
- The final model should report accuracy, precision, recall, F1-score, and ROC-AUC where appropriate.
- Recall should be monitored carefully because missing extended-wait cases may be more harmful than flagging too many cases.
- Class imbalance should be discussed, especially because the 4-hour wait target has relatively few positive cases.
- Feature importance or model coefficients should be explained in plain language.
- The model should be presented as an analytical prototype, not a clinical or operational automation system.

5.2 Final Scope Checks

- Confirm the project title, README, dashboard, SQL scripts, and reports use ED patient flow language.
- Confirm unsupported department-comparison claims have been removed.
- Confirm readmission analysis is excluded unless a valid readmission field is added from another approved dataset.
- Confirm recommendations are tied to evidence from the NHAMCS ED dataset.

6. Team Plan / Resource Plan

6.1 Team Roles

Team member	Main role	Responsibilities
Owen	Python, machine learning, dashboard, final analytics story	Python EDA, visualisations, feature engineering, model training, model evaluation, Streamlit dashboard, final insight storytelling.
Nimi	SQL, data modelling, KPI reporting, database documentation, business question alignment	SQL schema, data loading, SQL cleaning, KPI queries, SQL views, data dictionary, SQL documentation, SQL-backed business insights.

6.2 Contribution Plan

Work package	Lead	Support	Evidence on GitHub
Project specification and ED scope	Owen	Nimi	README.md and project planning notes.
Business questions and KPI mapping	Nimi	Owen	README business questions section and sql/04_kpi_queries.sql.
Data management plan and ethics	Nimi	Owen	docs/dataset_selection.md and README section.
SQL schema and cleaning	Nimi	Owen	sql/01_create_tables.sql and sql/03_cleaning_queries.sql.
SQL KPI queries and views	Nimi	Owen	sql/04_kpi_queries.sql and sql/05_views_for_python.sql.
Python EDA	Owen	Nimi	notebooks/01_python_eda.ipynb.
Machine learning	Owen	Nimi	notebooks/02_ml_long_wait_prediction.ipynb and model outputs.
Dashboard	Owen + Nimi	Owen + Nimi	dashboard/app.py with SQL insights included.
Executive summary and business insights	Owen + Nimi	Owen + Nimi	reports/executive_summary.md and reports/business_insights.md.
Final screenshots and repository cleanup	Owen + Nimi	Owen + Nimi	outputs/figures and final README.md.

6.3 Team Meetings / Group Sessions

Meeting type	Frequency	Purpose	Suggested day
Planning session	Weekly	Agree weekly goals, review previous milestone, assign tasks, and discuss blockers.	Sunday
Milestone review	Weekly	Check whether the weekly milestone has been met and update GitHub Issues.	Sunday
Progress check	Weekly	Review commits, check SQL/Python integration, and update issues.	Midweek / flexible
Working session	As needed	Pair work on integration points such as SQL views feeding Python and dashboard outputs.	Flexible
Final review	Once before submission	Test dashboard, check README, verify reports, and confirm deliverables.	Final week

6.4 Team Communication

- Use WhatsApp for quick communication and meeting reminders.
- Use GitHub Issues to record tasks, bugs, and review comments.
- Use pull requests for merging major SQL, Python, dashboard, and report changes.
- Use reports/business_insights.md to collect management-focused insights as the project develops.
- Use reports/executive_summary.md to summarise the final story for a non-technical hospital management audience.
- At the end of each week, each person should commit visible evidence of their work to GitHub before the Sunday review.